



NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
Third Semester B.Tech (E&E) (CBCS) Degree Examinations
December 2023

EE2104-1 – GENERATION TRANSMISSION AND DISTRIBUTION

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

1. A hydroelectric power station is commonly installed at
A) desert areas B) hilly areas
C) Swamps D) grasslands
2. Penstock is used as a conduit between
A) The steam chest and the turbine in a thermal station B) The dam and the turbine in a hydro station
C) The turbine and the discharge drain D) The turbine and the tail race
3. Nuclear reactors generally employ
A) Fission B) Fusion
C) Both fission and fusion D) Steam production
4. The major heat loss in a steam power station occurs in
A) Penstock B) spillways
C) Condenser D) reactor
5. Demand factor is the ratio of
A) Connected load to maximum demand B) Maximum demand to connected load
C) Average load to maximum demand D) Maximum demand to average load
6. Maximum demand of a generating station is 100 MW, a load factor is 65% Find the average demand
A) 65 MW B) 6.5 kW
C) 1.54 MW D) 65 kW
7. Load curve is useful in deciding
A) Operating schedule of generating units B) Sizes of generating units
C) Total installed capacity of the plant D) All of the above
8. Most efficient plants are normally used as
A) Peak load plants B) Base load plants
C) Both peak and base load plants D) None of these
9. The sag of a transmission line depends on
A) Length of the conductor B) Weight per unit length of the conductor
C) Conductor tension D) All of these
10. Span length is the ----- distance between the two towers
A) Shortest B) largest
C) Average D) None of these
11. Why is high voltage transmission line transposed?
A) Corona losses can be minimized B) Computation of inductance becomes easier
C) Voltage drops in the lines can be minimized D) Phase voltage imbalances can be minimized
12. In the case of suspension-type insulators, the string efficiency can be improved by
A) using a longer cross arm B) using a guard ring
C) grading the insulator discs D) all of these
13. The transmission lines are classified as short, medium and long lines based on?
A) Nominal voltage of the line B) Physical length of the line
C) Wavelength of the line D) Power transmitted over the line

14. What happens if the separation between the three phases of the transmission line is increased?
 A) The inductance will increase and capacitance will remain unchanged
 B) Both inductance and capacitance will decrease
 C) Inductance will increase and capacitance will decrease
 D) Inductance will decrease and capacitance will increase
15. The capacitance effect can be neglected in which type of transmission lines?
 A) Short transmission lines
 B) Medium transmission lines
 C) Long transmission lines
 D) All of these
16. Identify the characteristics of the Ferranti effect:
 (i) Flow of heavy current
 (ii) It occurs at unloaded condition
 (iii) Rise in the receiving end voltage
 A) (i) and (ii)
 B) (ii) and (iii)
 C) (i) and (iii)
 D) (i), (ii) and (iii)
17. What does capacitance grading of cables mean?
 A) Use of dielectrics in different concentrations
 B) Introduction of capacitance at various lengths of cable to counter the effect of inductance
 C) Use of dielectrics of different permittivity's
 D) Grading according to capacitance per km length of the cable
18. Name the wire which connects the distributor to the consumer terminals
 A) Distributors
 B) Service mains
 C) Feeders
 D) All of these
19. Grading of cables
 A) Reduces insulation cost and increases current rating
 B) Reduces insulation cost but decreases current rating
 C) Increases insulation cost and increases current rating
 D) decreases insulation cost and decreases current rating
20. FACTS devices used in
 A) AC transmission
 B) DC transmission
 C) Generation
 D) None of these

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I

- | | Marks | BT* | CO* | PO* |
|--|-------|-----|-----|-----|
| 1. a) Draw and explain the schematic diagram of Thermal power plant. | 8 | | | |
| b) A generating station has the following data:
Installed capacity = 500 MW
Capacity factor = 45%
Annual load factor = 60%
Annual cost of fuel, oil etc. = Rs 10×10^7
Capital cost = Rs 10^9
Annual interest and depreciation = 15%
Calculate: i) the minimum reserve capacity of the station
ii) the cost per kWh generated. | 8 | L3 | 2 | 1,2 |
| 2. a) Mention the pros and cons of nuclear power generation. | 4 | L1 | 1 | 1,2 |
| b) Mention the points to be considered for the selection of site in case of Hydro Power generation. | 4 | L2 | 1 | 1,2 |
| c) What is Tariff? Explain two-part tariff, block rate tariff and power factor tariff. | 8 | L2 | 2 | 1,2 |
| 3. a) Draw and explain the general arrangement and operation of hydroelectric power plant. | 8 | L2 | 1 | 1,2 |
| b) A generating station supplied the following loads: 150MW, 120MW, 85MW, 60MW and 5MW. The station has a maximum demand of 220MW. The annual load factor of the station is 48%. Calculate:
i) The number of units supplied annually
ii) The diversity factor and
iii) The demand factor. | 4 | L3 | 2 | 1, |
| c) Write the advantages of high voltage transmission. | 4 | L1 | 2 | 1, |

Unit – II

- | | | | | | |
|-------|---|---|----|---|-----|
| 4. a) | Briefly explain the potential distribution over a string of suspension insulators. | 8 | L2 | 3 | 1,2 |
| b) | Derive the expression for sending end voltage and current by modelling the medium transmission line using nominal π method. | 8 | L2 | 4 | 1,2 |
| 5. a) | Derive the expression for sag when conductors are suspended at unequal levels of support. | 8 | L2 | 3 | 1,2 |
| b) | Determine the ABCD constants for a medium transmission line for nominal T-representation and prove that $AD-BC = 1$. | 8 | L2 | 4 | 1,2 |
| 6. a) | Derive the expression for inductance of a single-phase two-wire line. | 4 | L2 | 3 | 1,2 |
| b) | Explain the different methods of testing of insulators. | 4 | L2 | 3 | 1,2 |
| c) | Determine A, B, C, D constants for a 3 phase 50 Hz long transmission line 200 km long having the following distributed parameters:
Inductance $= 120 \times 10^{-3}$ H/km
Capacitance $= 8 \times 10^{-9}$ F/km
Resistance $= 0.15 \Omega/\text{km}$ | 8 | L3 | 4 | 1,2 |

Unit – III

- | | | | | | |
|-------|--|---|----|---|-----|
| 7. a) | Explain inter sheath grading of underground cables and derive the expression for stress. | 8 | L2 | 5 | 1,2 |
| b) | A 33kV single core cable has a conductor diameter of 2 cm and a sheath of inside diameter 3 cm, find maximum and minimum stress in the insulation. | 4 | L3 | 5 | 1,2 |
| c) | State the advantages (any 4 points) and disadvantages (any 4 points) of HVDC transmission. | 4 | L1 | 5 | 1,2 |
| 8. a) | Explain capacitance grading of underground cables and derive the expression for stress. | 8 | L2 | 5 | 1,2 |
| b) | Explain radial and ring main system of ac distribution. | 8 | L2 | 5 | 1,2 |

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
